THE UNIVERSITY OF DA NANG
VNUK INSTITUTE FOR RESEARCH AND EXECUTIVE EDUCATION

Introduction to Human-Computer Interaction
Final Project Report

Opportunities

A note-taking app for mobile and web

Capture fast. Organize clearly. Find it again.

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Submitted [submission date]

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Part 1: Essay

I. Brief

Background

- People keep notes in many places at once: a phone, a laptop, a paper notebook, chat apps. When they need one note, they lose time searching across all of them.
- Handwriting, screenshots and scanned pages cannot be searched, and typing is slow when someone is busy or on the move.
- People also forget deadlines, lose work when offline, and give up on tools that demand Markdown or LaTeX knowledge up front.
- **Opportunities** brings text, voice, photos and scans, checklists, tables and formulas into one note app. The guiding question: how might we help people capture, organize and find notes again, without having to learn Markdown first?

A. Project

Opportunities: a Markdown-first, offline-first note-taking application for mobile and web.

B. Brand

Opportunities. The logo is a violet cube (#A882FF) on charcoal.

C. Client

This is a course project with no external client. It is designed for students, tutors and knowledge workers who take notes every day.

D. Project deliverables

- Brand introduction and logo
- Research: scenarios, two personas, sympathy map
- Flow charts and UX wireframes
- Design system with Dark and Light themes
- Interactive mobile prototype and web prototype in Figma
- Test cases and a usability test plan
- This report and presentation slides

E. Budget

N/A (academic project).

F. Brand background

- Created in 2026 by Trần Ngọc Quân and Ngô Hoàng Nhật Khang for the VNUK Human-Computer Interaction module.
- Motto: “Capture fast. Organize clearly. Find it again.”
- Visual language: charcoal neutrals with one violet accent, the Inter typeface and line icons, inspired by Obsidian.

G. Target audience

- **Primary:** students and tutors with everyday tech skills, who capture on a phone and need to find things quickly.
- **Secondary:** power users such as developers and researchers, who write Markdown and LaTeX and want to own their data.

H. Coverage

Vietnamese-speaking users first, with Vietnamese as the default interface language and English as the second. Target platforms are iOS 16+, Android 10+ and web (Chrome, Edge, Safari, Firefox).

I. Timing

- 27 October 2026: version 5, full redesign of the mobile prototype.
- 1 November 2026: version 6, lecturer feedback applied.
- Final submission: [submission date].

J. Description

The core flow is capture, organize, find, act on reminders, then share or export. One editor serves both people who only tap buttons and people who type Markdown or LaTeX. Mobile navigation has five items: Notes, Review, Create (+), Reminders and Profile.

K. Objective

- Give one place for every kind of note, so nothing is scattered across apps.
- Make any note easy to find again, including text inside photos and scans.
- Keep the app simple for casual users while staying powerful for Markdown and LaTeX users.
- Protect user data through auto-save, offline work, a 30-day trash and version history.

II. Analyze

A. Scenarios

Scenario 1: a day with Trần Minh Thư. Each step maps to a real screen in the mobile prototype.

Step	Situation	What Thư does	Screens
1	07:55, on the bus	Taps + then Voice. Speech is written into a meeting note as she talks.	2.1, 3.2
2	09:40, in class	Scans the whiteboard. The app detects the edges and captures two pages. She checks the recognized text and fixes two flagged words.	3.3, 3.4
3	09:45, writing a formula	Types \fr and picks Fraction from the suggestions, without leaving the editor.	2.8
4	12:30, no network	A banner says she is offline. Her note stays on the device and changes wait to sync.	7.1
5	17:30, tuition reminder	A lock screen notification appears. She opens Reminders and marks room 3 as collected.	5.3, 5.8
6	20:30, on her laptop	A note was edited in two places. She compares the versions and keeps both, so no line is lost.	7.2, 7.4
7	21:00, sharing with the group	Opens the note ⋯ menu, confirms a public link for 7 days, and sends the QR code.	2.9, 6.1, 6.2
8	23:00, revising for the exam	Searches “persona” and finds it inside this morning’s scan, reads in Focus mode, then reviews flashcards.	4.2, 2.31, 2.11, 11.1

Scenario 2: Phạm Đức Anh writes a thesis note.

Step	Situation	What Đức Anh does	Screens
1	Starting a thesis note	Taps + then Note, and switches to Markdown source from the mode box.	2.1, 2.2, 2.31, 2.5
2	Writing an equation	Types LaTeX with autocomplete and sees it rendered in place.	2.8
3	Linking ideas	Types [[to link a paper summary, using the suggestion list.	2.7
4	Grouping thesis notes	Opens the ☰ menu, manages folders, and creates a smart folder filtered by tag.	1.28, 4.4, 12.1, 12.4
5	Sending to the advisor	Exports the note as PDF with the chosen page size and text size.	6.3, 6.9
6	Going back to an old draft	Opens version history and restores an earlier version without losing the current one.	7.3, 7.7

B. Persona

We created two personas: a primary everyday user and a power user. Together they cover the range from people who only tap buttons to people who type Markdown.

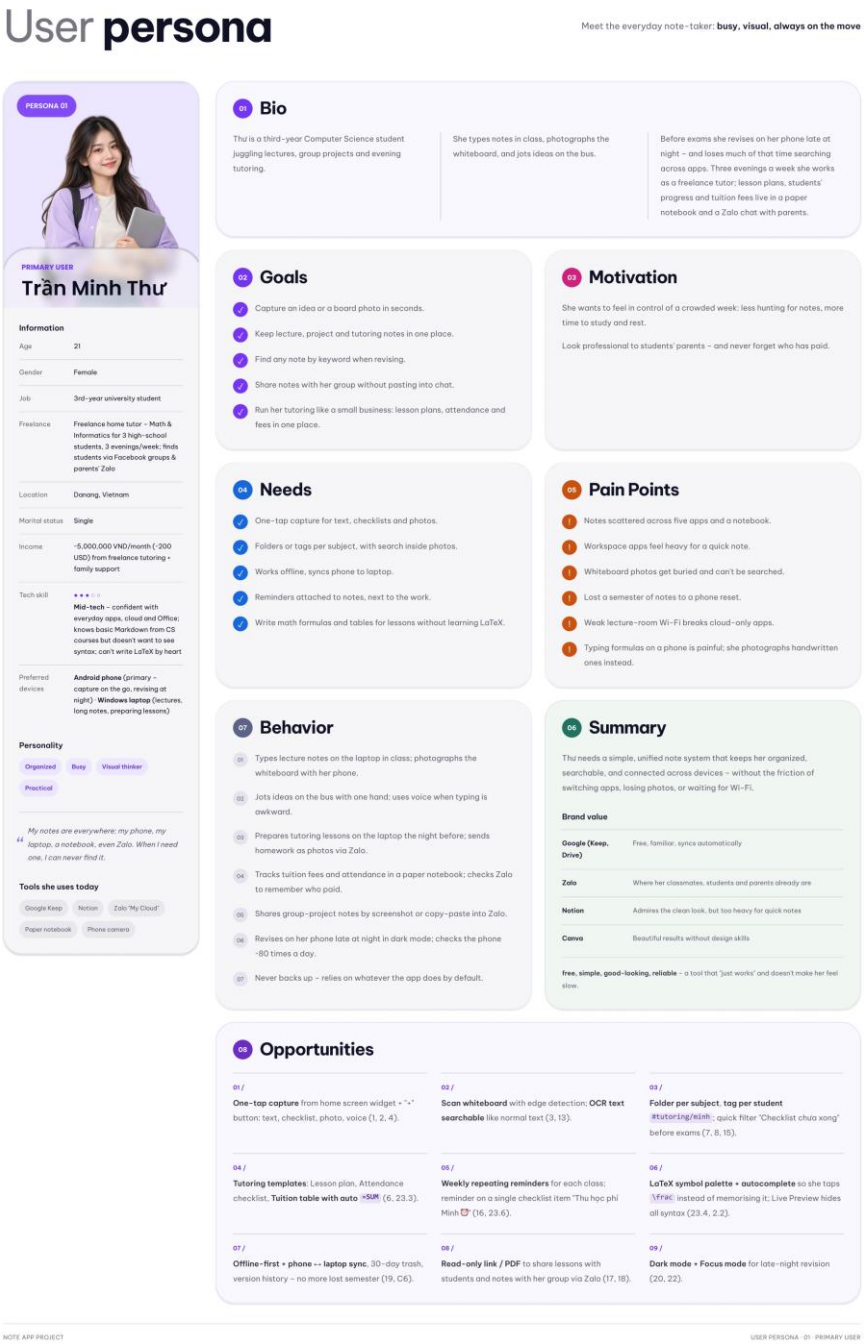


Figure 1. Persona 1: Trần Minh Thư, primary user

Trần Minh Thư, 21	**Summary**
Goals	Capture an idea or board photo in seconds. Keep lecture, project and tutoring notes in one place. Find any note by keyword when revising.
Pain points	Notes scattered across five apps and a notebook. Whiteboard photos cannot be searched. Weak lecture-room Wi-Fi breaks cloud-only apps. Typing formulas on a phone is painful.
What Opportunities gives her	One-tap capture, scan with searchable text, offline work with sync, reminders next to the work, tutoring templates.

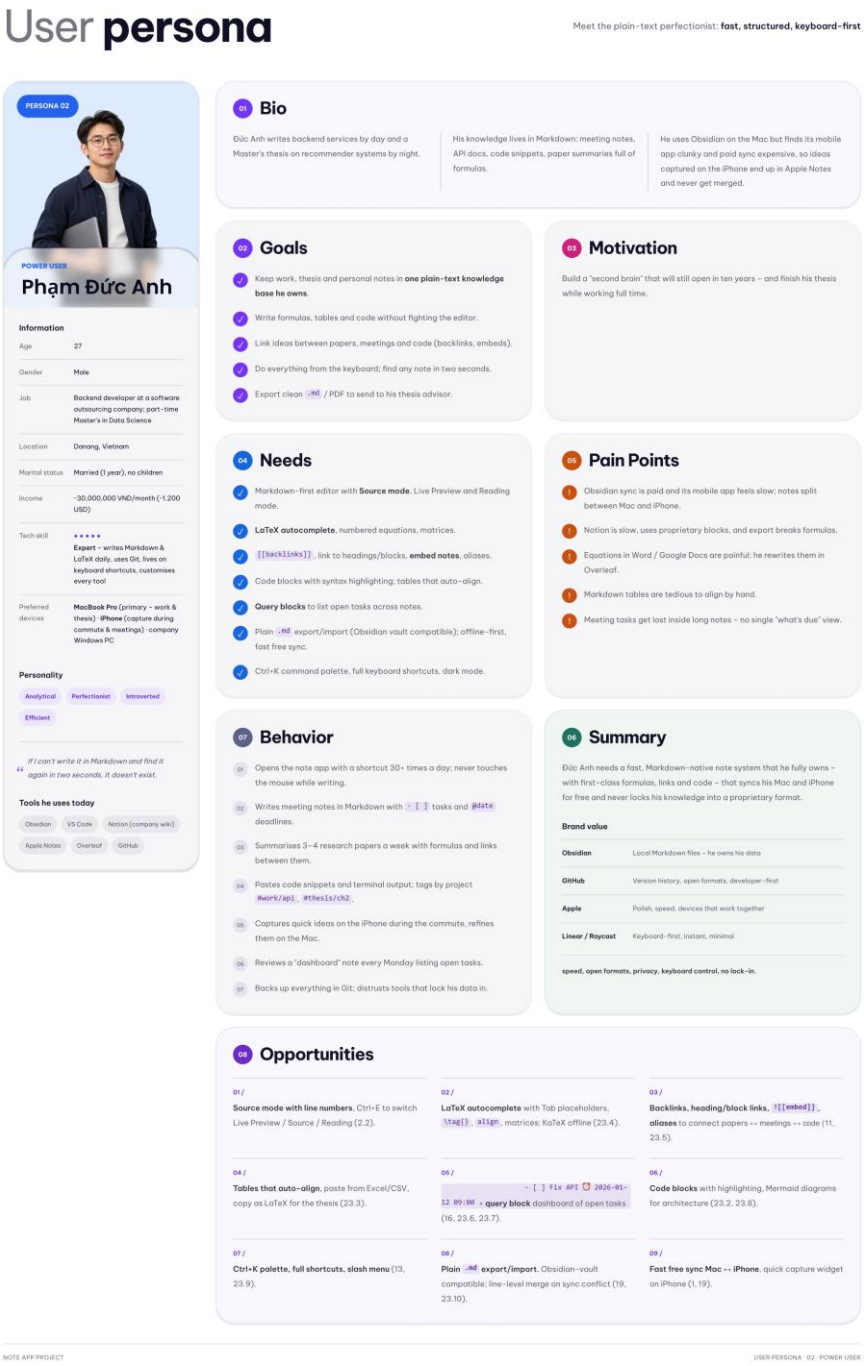


Figure 2. Persona 2: Phạm Đức Anh, power user

Phạm Đức Anh, 27	**Summary**
Goals	Write formulas, tables and code without fighting the editor. Link ideas with backlinks. Find any note in two seconds. Export clean Markdown or PDF for his advisor.
Pain points	Paid sync and a slow mobile app in his current tool. Export breaks formulas in other tools. Meeting tasks get lost inside long notes.
What Opportunities gives him	Markdown source mode, LaTeX autocomplete, wiki-links, plain .md export, smart folders, keyboard shortcuts on web.

C. Sympathy map

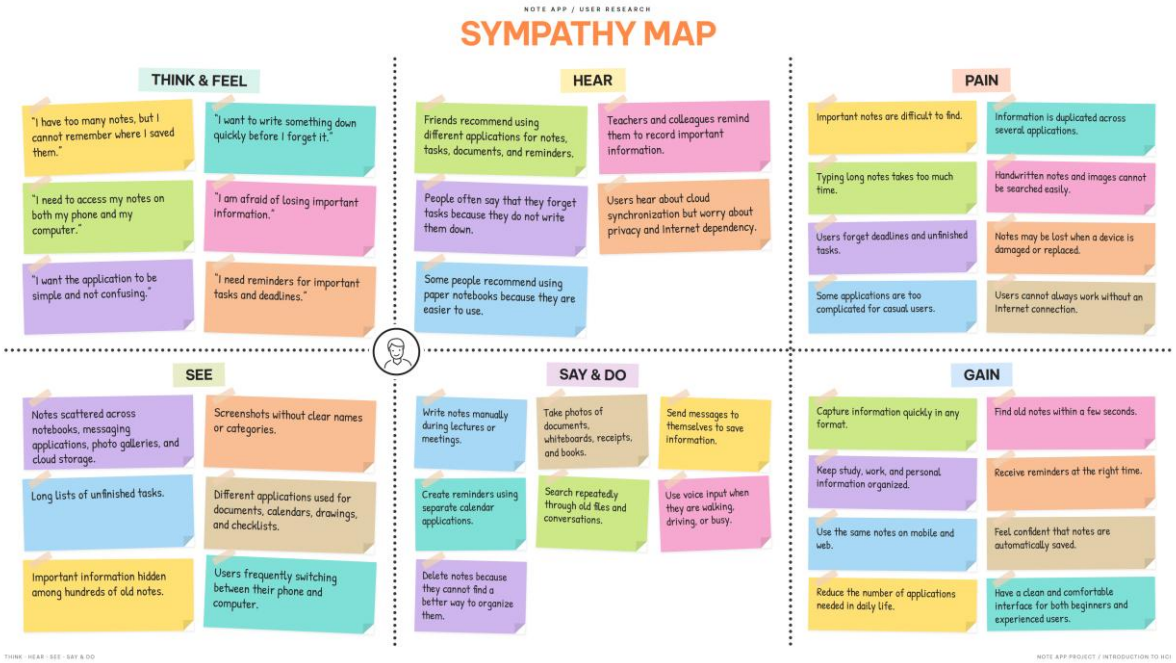


Figure 3. Sympathy map

Key insights from the map:

- Important notes are hard to find, and people switch between many apps to look for them.
- Typing long notes takes too much time, so people want to capture in any format.
- People forget deadlines and unfinished tasks, so reminders must sit next to the work.
- Some apps are too complicated for casual users, so advanced features must stay out of the way.

D. Feature list

Functional requirements

Area	Features
Getting started	Sign in, sign up, password recovery, guided tour, sample note, help and syntax cheat sheet.
Browse	Grid, List, Calendar, Kanban and link graph views. Sort, filter and quick chips. Pin, move, color, remind, duplicate, share, archive, delete. Multi-select.
Write	Preview, Markdown source, Read only and Focus modes. Slash menu, wiki-links, formula suggestions. Note color, tags and custom properties. Find in note. Lock note.
Capture	Note, checklist, voice, photo, scan, drawing, template. Scan images or PDFs from the device. Text recognition with review. Attach PDF or DOCX.
Find and organize	Search including text inside photos. Nested folders, smart folders, tags, attachment library.
Remind and revise	Reminders by due date, snooze, notifications, quiet hours. Flashcard review with spaced repetition.
Share and sync	Public link with QR, revoke. Export PDF and Markdown. Offline work, conflict resolution, version history.
Safety and settings	Undo, 30-day trash, archive. Dark and Light themes, text size, paper style, language, note lock, devices, import and export.
Templates	Built-in templates, save a note as a template, edit, rename, duplicate, delete.

Non-functional requirements (design targets)

Category	Target
Performance	Open the app in 2 s or less. Open a note in 0.5 s or less. Search in 300 ms or less across 1,000 notes.
Offline	Everything works offline except first sign-in and creating share links.
Data integrity	Continuous auto-save, 30-day trash, no data loss on sudden close, full export as ZIP.
Security	HTTPS, passwords of 8 or more characters, revocable random share links, remote sign-out.
Accessibility	Touch targets of 48×48 px or more, WCAG AA contrast, labels or tooltips on icons, screen reader support, full keyboard use on web, never rely on color alone.
Compatibility	iOS 16+, Android 10+, the latest two versions of Chrome, Edge, Safari and Firefox. Vietnamese and English.

III. Produce

A. Flow chart

Diagram labels are in Vietnamese, matching the app interface. The full editable set is in the project’s draw.io file.

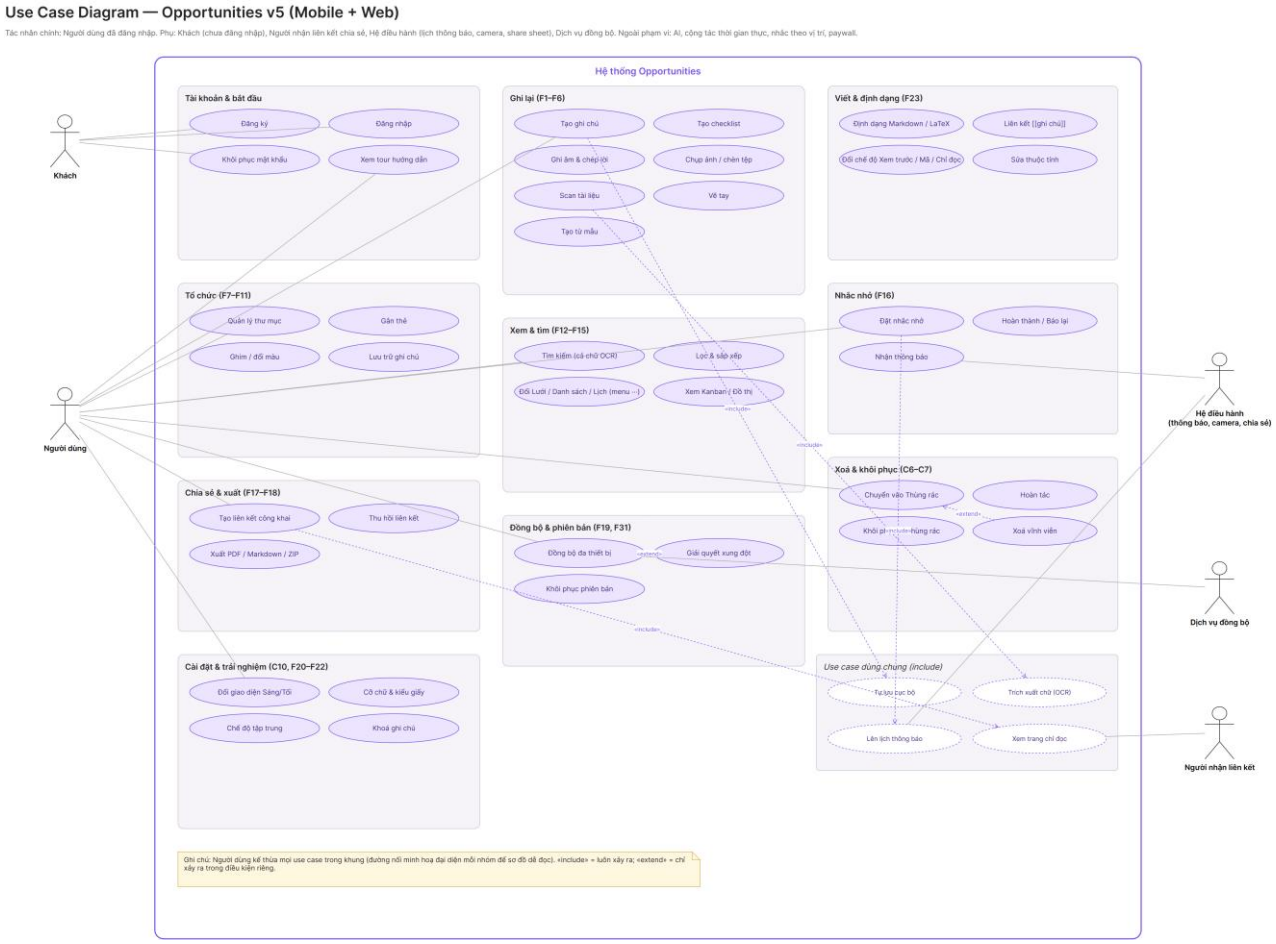


Figure 4. Use case diagram

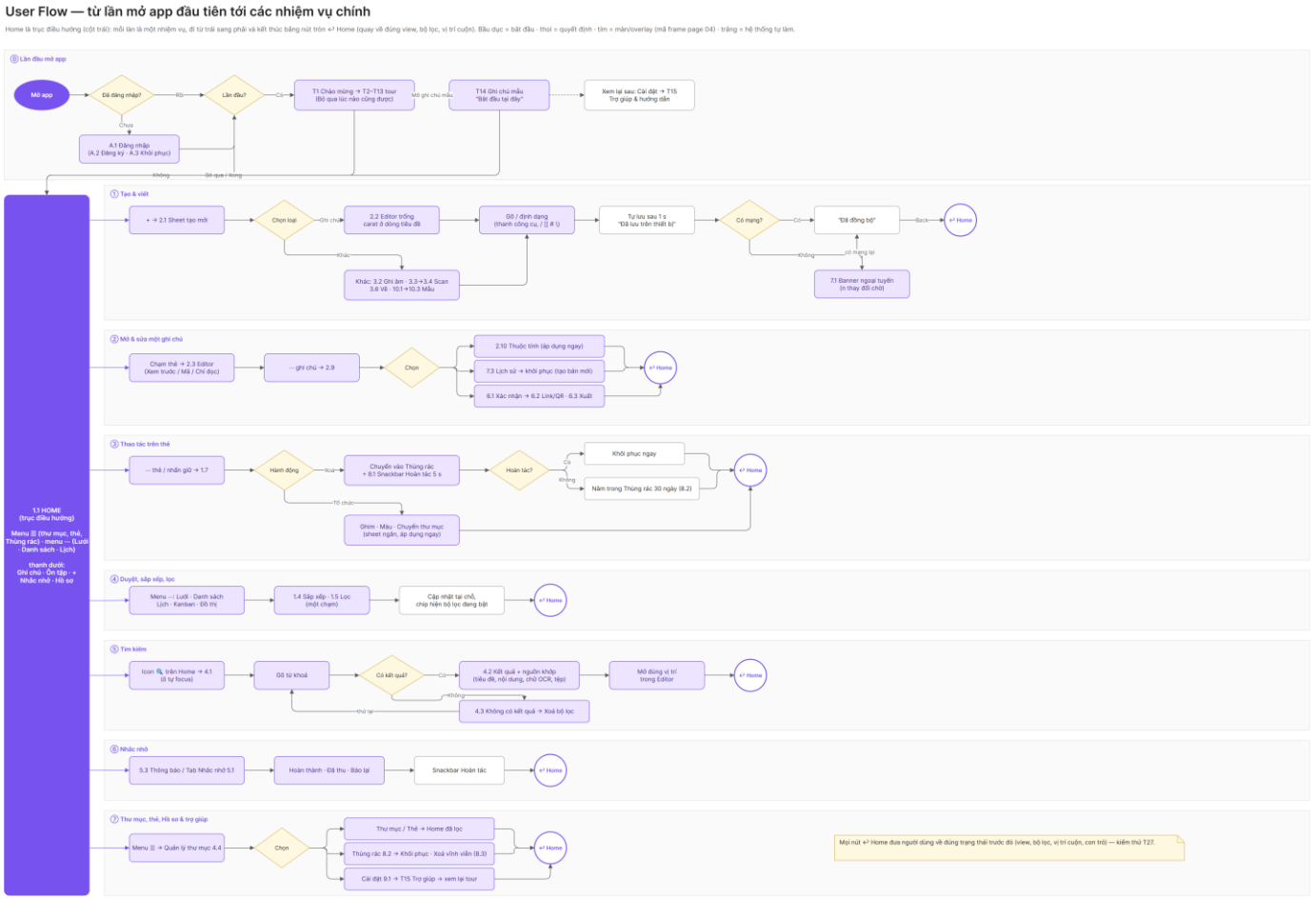


Figure 5. User flow, from first open to each main task

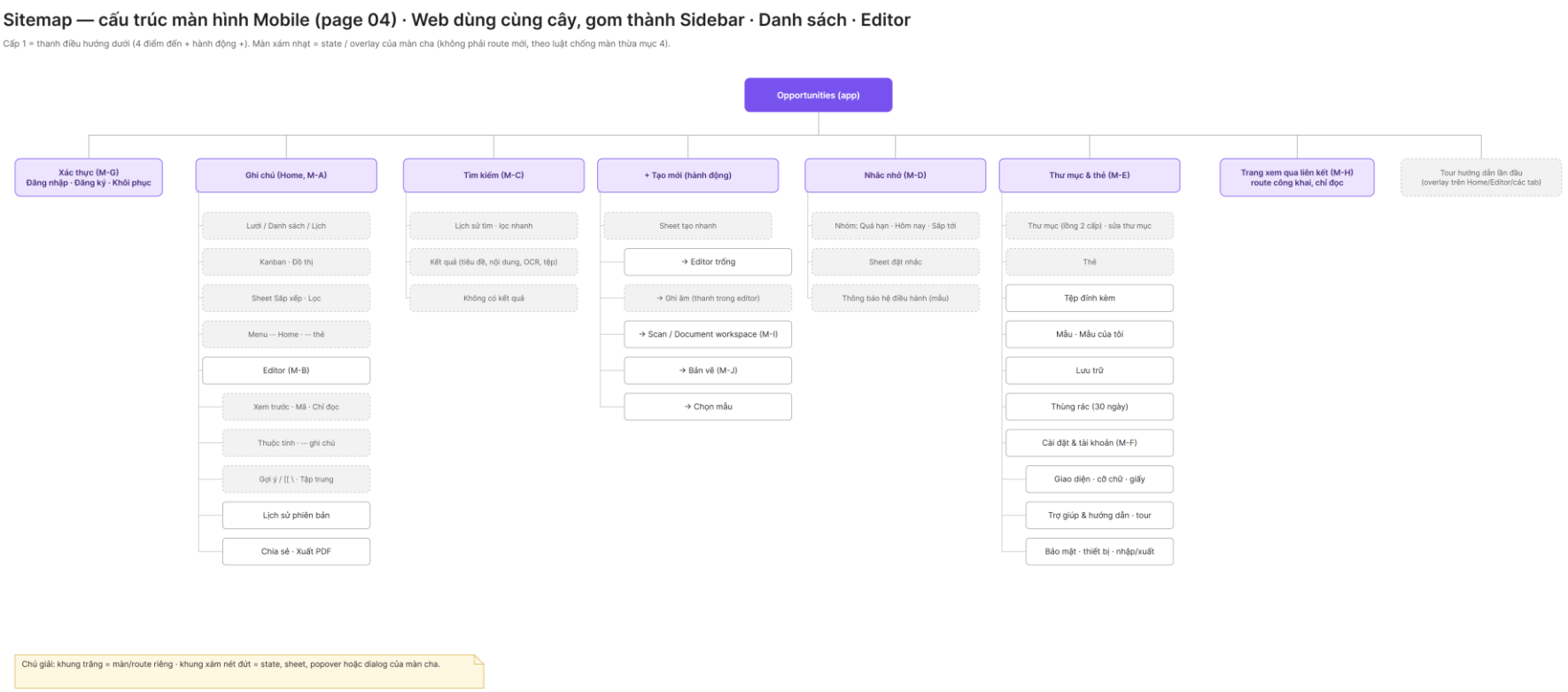


Figure 6. Sitemap of the mobile app

Information Architecture — mô hình nội dung · cách tổ chức · điều hướng · nhãn · tìm kiếm

Theo 4 hệ thống IA (Rosenfeld & Morville): Organization, Labeling, Navigation, Search — đặt trên mô hình dữ liệu Markdown-first (mục 3 spec).

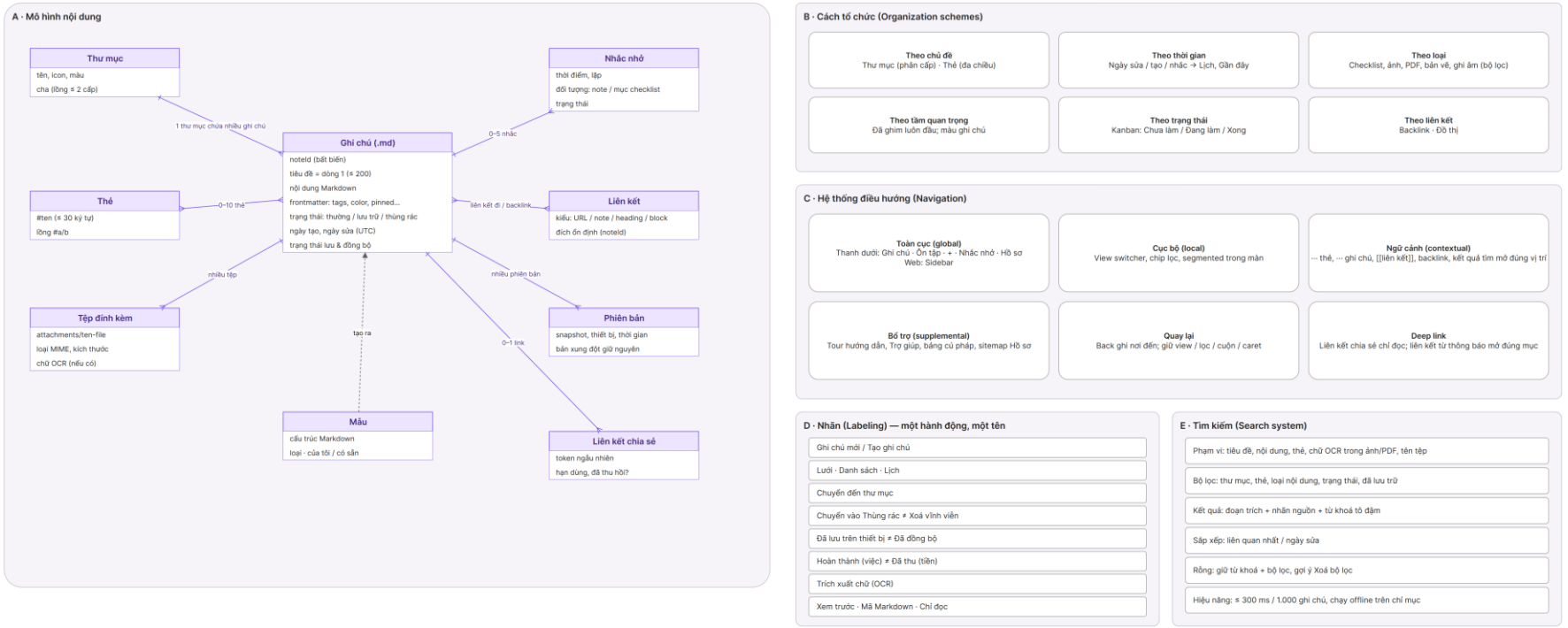


Figure 7. Information architecture

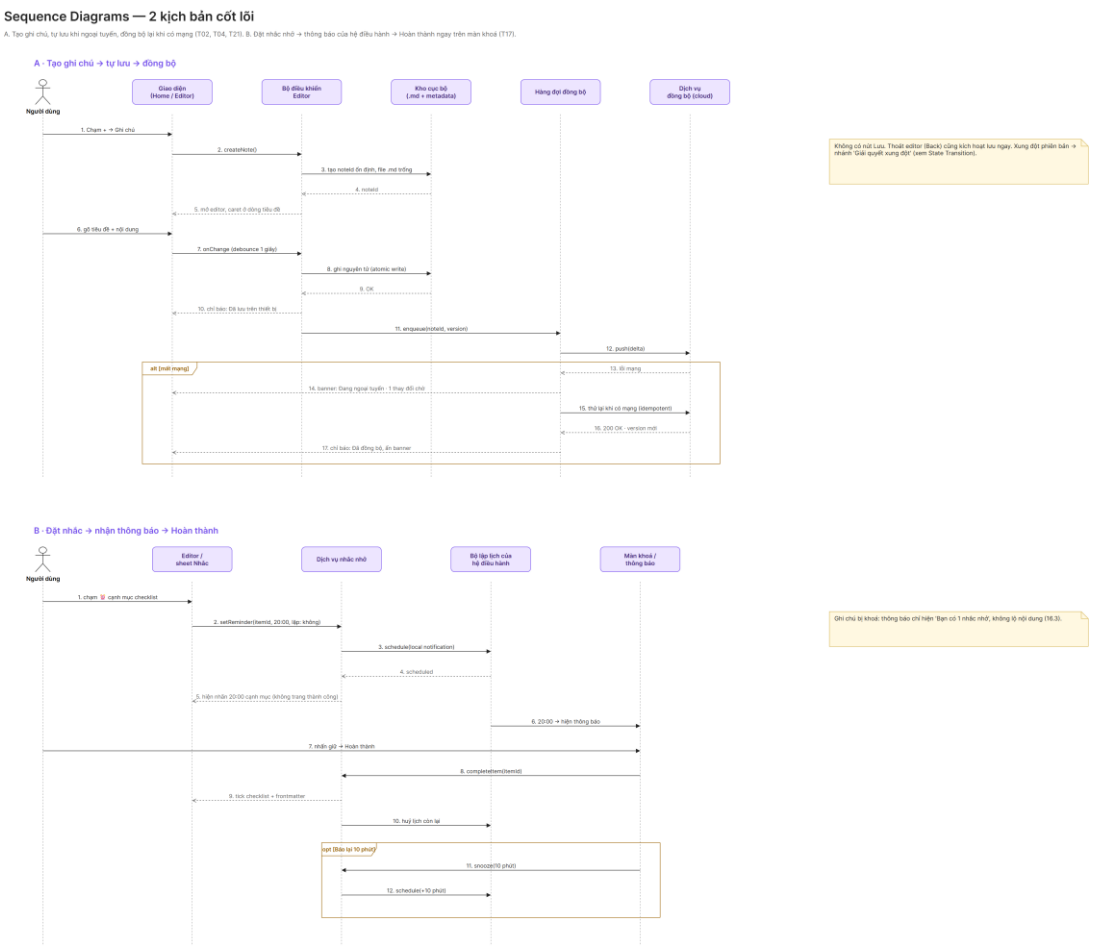


Figure 8. Sequence diagrams for two core scenarios

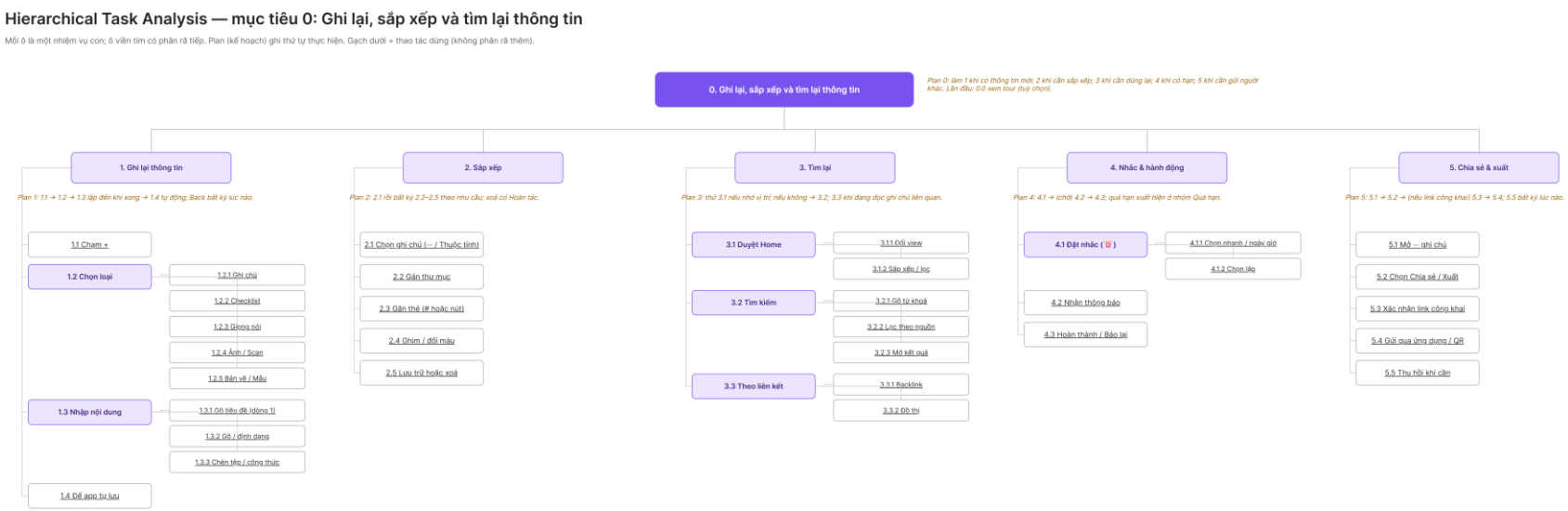


Figure 9. Hierarchical task analysis

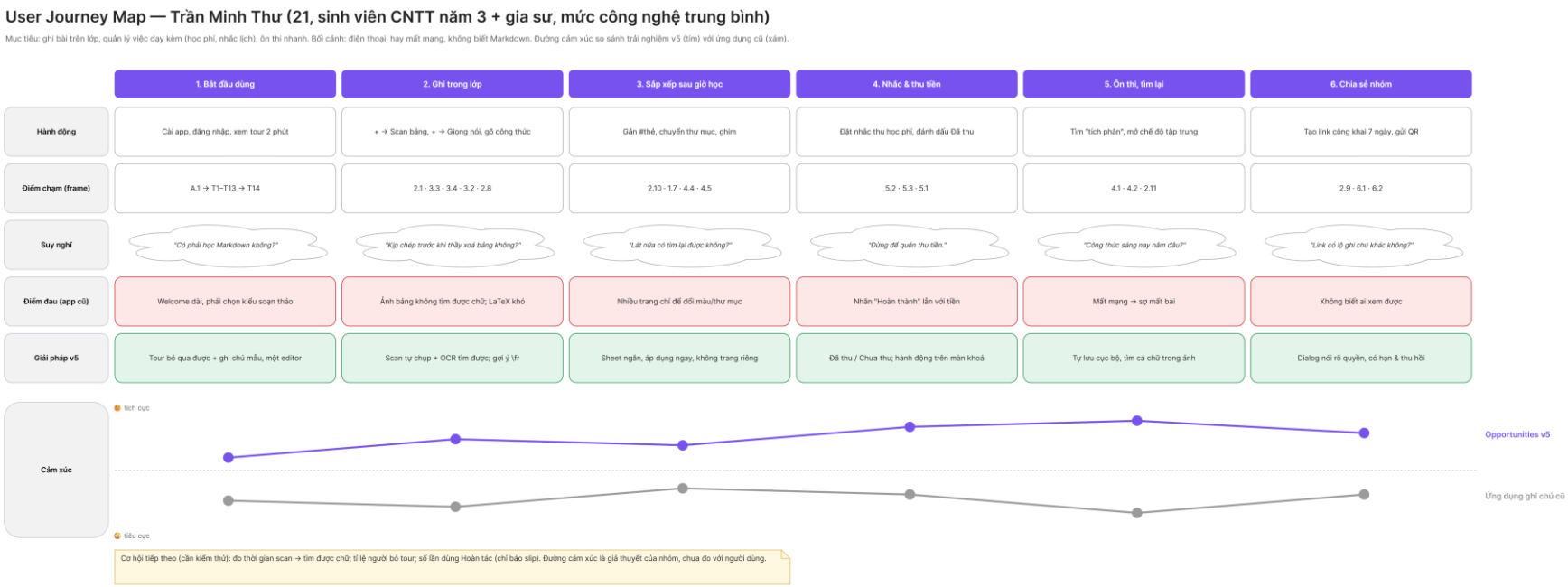


Figure 10. User journey map of Trần Minh Thư

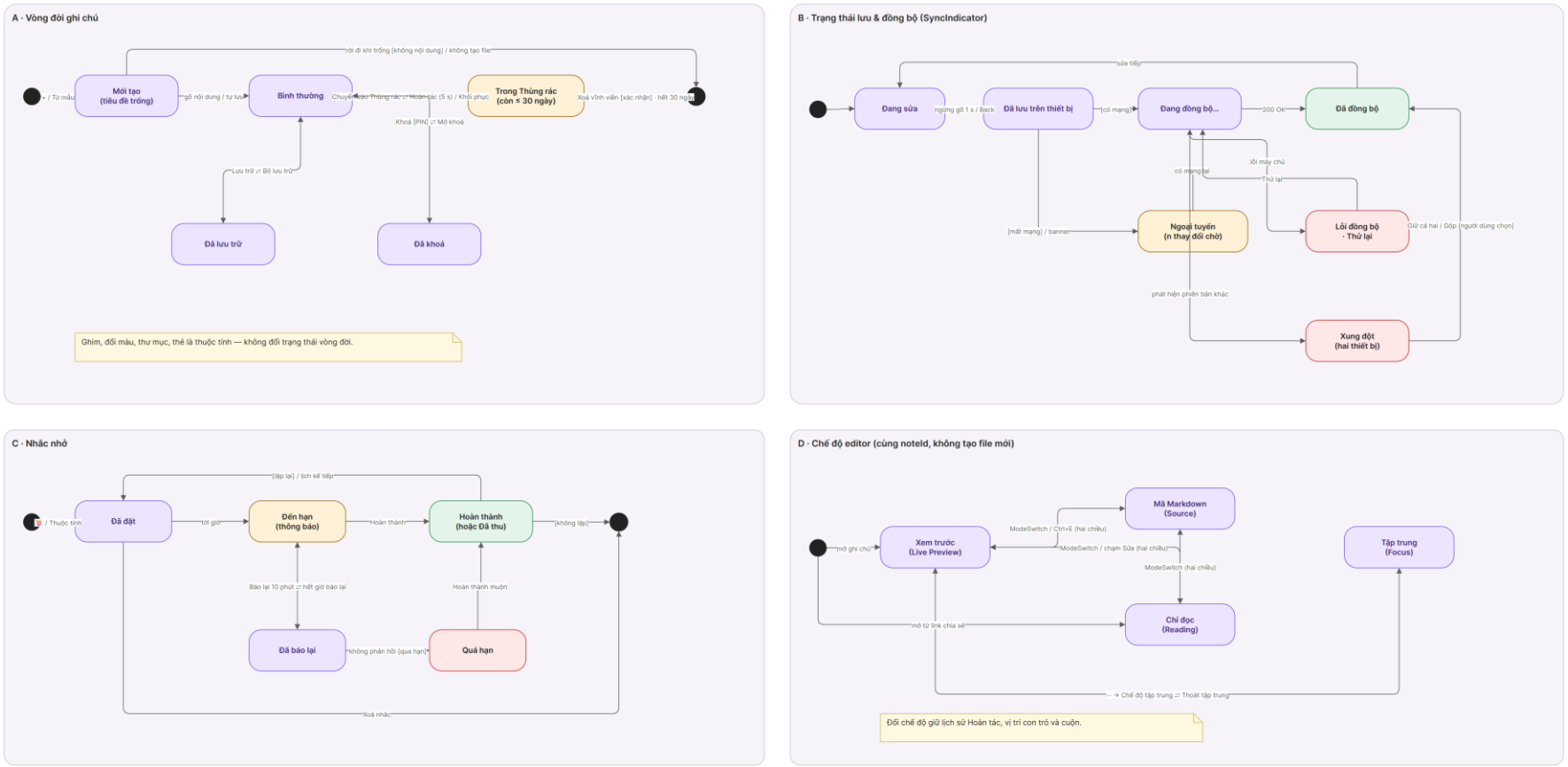


Figure 11. State transition diagrams

B. UX wireframe

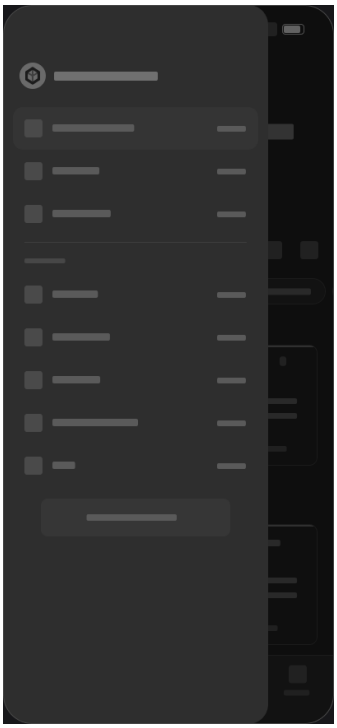
Wireframes use gray blocks only, with no icons or real data, so the review stays on structure. Every screen in the main sections has one, each with a note on the HCI principle and the design decision. Below are the key screens, with the principle first and the decision second.



1.1 Home

Recognition over recall, minimal design

Spacious header, ☰, search and ⋯ in one row, quick chips, pinned notes first



1.28 Side menu

Grouping by purpose

Secondary navigation moves into ☰ so Home stays clean



2.1 Create sheet

Fewer steps, large targets (Fitts's law)

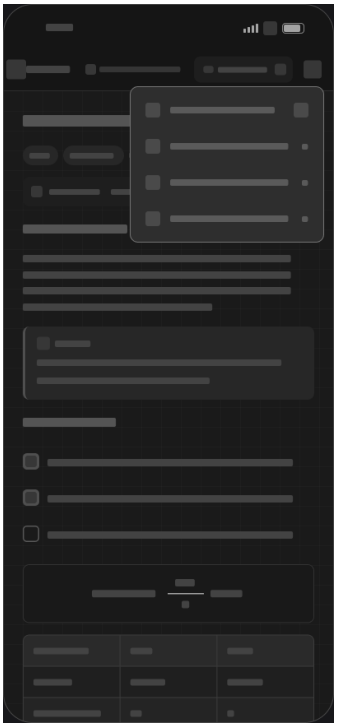
+ is an action; Note is the main button, other tiles go straight to the tool



2.3 Editor

Content first

One editor; Markdown marks show only on the line with the caret



2.31 Mode dropdown

Grouping, recognition over recall

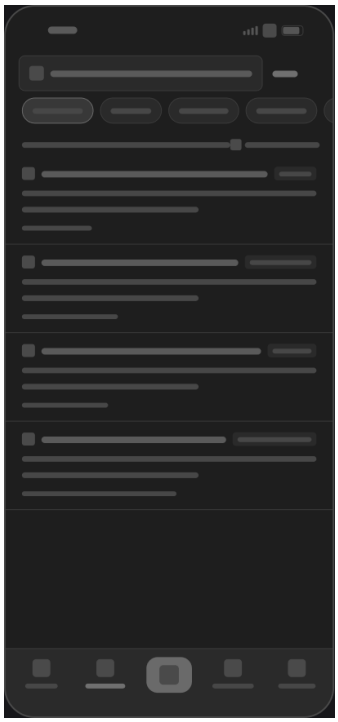
All four modes sit together in the mode box; a tick marks the current one



3.4 Text review

Error prevention

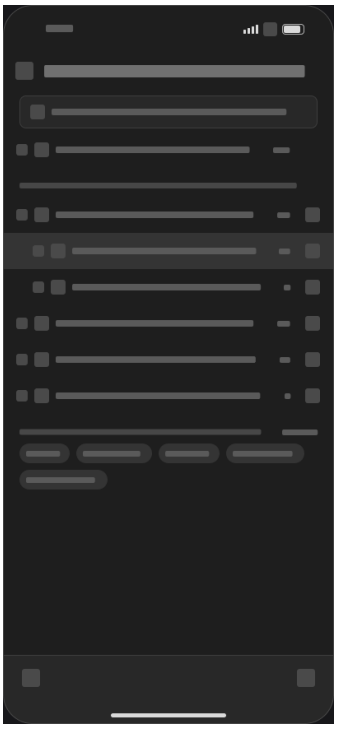
Review and fix recognized text; doubtful words are flagged; the original is kept



4.2 Search results

Visibility of why it matched

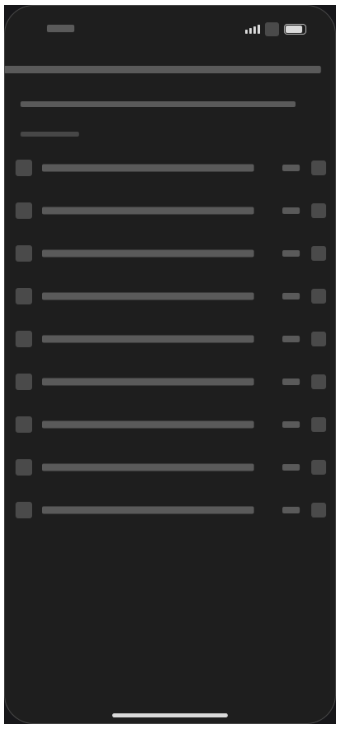
Labels show the match source: title, content, text in photo, file



4.4 Folders

Clear hierarchy

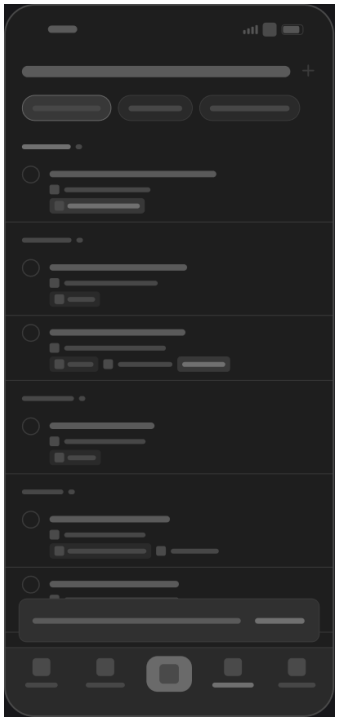
Own screen like Apple Notes: search, nested folders, tags, new folder and note



12.1 Smart folder filters

Safe defaults

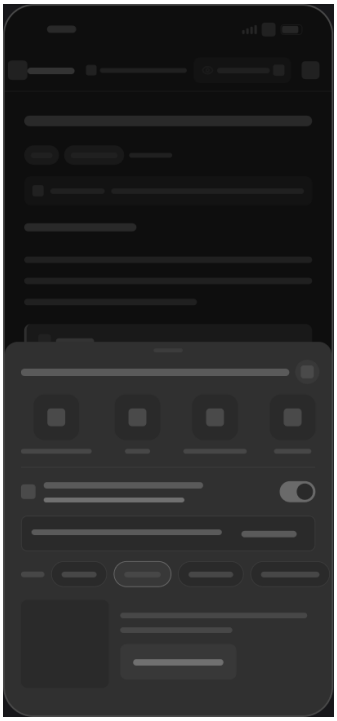
Filter by tag, date or type; every filter starts off



5.1 Reminders

Grouping by time

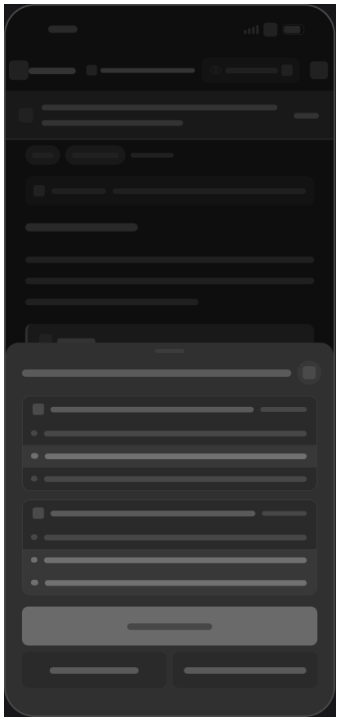
Overdue, today, tomorrow, upcoming, with wording that fits the task



6.2 Share

User control

Link has an expiry and can be revoked; QR matches the link



7.2 Conflict

No data loss, safe default

Compare two versions; Keep both is the default



9.1 Profile

One grouped list

All settings on one scrolling screen, no separate page for a single switch



11.1 Review

One main action

Cards due today and a single Start button

IV. Test cases

These test cases describe the expected behavior of the design, and the prototype implements each path. They have not yet been run with real users, so no pass or fail results are reported here.

ID and feature	Precondition	Steps	Expected result
TC-01 Sign in	Valid test account	1. Open the app. 2. Enter email and password. 3. Tap Sign in.	Home opens with no welcome pages.
TC-02 Sign-up validation	On sign-up screen	1. Enter an invalid password. 2. Leave the field.	An inline error explains the problem before submit (A.2).
TC-03 Create a note	Signed in	1. Tap +. 2. Tap Note.	Editor opens directly with the caret on the title (2.2).
TC-04 Auto-save	Note open	1. Type text. 2. Wait 1 second. 3. Go back and reopen.	Text is still there. No Save button is needed.
TC-05 Change view	On Home	1. Tap ⋯. 2. Tap List. 3. Tap ⋯ again.	Home shows List (1.2). The tick moves to List (1.6b).
TC-06 Editor modes	Note open	1. Tap the mode box. 2. Tap Markdown source. 3. Open ⋯.	Source view opens (2.5). The ⋯ menu lists no modes.
TC-07 Scan a document	Camera allowed	1. Tap +, then Scan. 2. Capture a page. 3. Review, then create.	A new note holds the scan and its text (2b.9).
TC-08 Scan from device	Image on the device	1. Tap +, then Scan from device. 2. Pick an image. 3. Review the text and create.	Review appears before the note is created (12.7).
TC-09 Search text in a photo	A scanned note exists	1. Tap search on Home. 2. Type a word from the scan.	The result opens the note and shows the match source (4.2).
TC-10 Smart folder	On Folders	1. Tap New folder. 2. Make it a smart folder. 3. Turn on a tag filter, tap Done.	The smart folder appears in the list with a confirmation (12.4).
TC-11 Folder actions	On Folders	1. Tap ⋯ on a folder. 2. Edit and save. 3. Tap ⋯, delete, confirm.	Each action ends on an updated list with a snackbar (4b.3, 4b.5).
TC-12 Reminder	On Reminders	1. Tap +. 2. Fill in and save. 3. Open it and mark done.	It appears in the list, then moves to completed (5.7, 5b.8).
TC-13 Flashcard review	On Review	1. Tap Start. 2. Rate each card. 3. Tap Back to Review.	Summary shows the ratings (11.5). Review shows nothing due (11.2).
TC-14 Share and revoke	Note open	1. Open ⋯, then Share. 2. Confirm the public link. 3. Revoke it.	Link and QR appear (6.2). After revoking, the state updates (6.8).
TC-15 Export PDF	Note open	1. Open ⋯, then Export. 2. Change page size. 3. Export.	Preview updates and the export is confirmed (6.10, 6.9).
TC-16 Offline and conflict	Same note edited on two devices	1. Reconnect. 2. Open the conflict. 3. Tap Keep both.	Both versions are kept, with no text lost (7.4).
TC-17 Delete and restore	Note exists	1. Delete the note. 2. Tap Undo, or open Trash from ☰. 3. Tap Restore.	The note returns to its original folder (8.5).

ID and feature	Precondition	Steps	Expected result
TC-18 Lock a note	Note open	1. Open ⋯, then Lock. 2. Confirm.	The note is locked (2.29). Its content is hidden from search, widgets and notifications.
TC-19 Use a template	On the create sheet	1. Tap Template. 2. Preview one. 3. Use it.	Editor opens with the template filled in (10.3).
TC-20 Settings and sign-out	On Profile	1. Change the language. 2. Scroll down. 3. Tap Sign out.	The new value shows at once (9.15). Sign out returns to A.1.

Usability test plan

Method: 5 participants per persona, think-aloud, and a SUS questionnaire after the tasks. Any task under 80% success goes into the next design iteration. These are planned targets, not measured results.

Task for the participant	Persona	Success criterion	Target
Save this whiteboard photo so you can find it later	Thư	Note saved with text	4 taps, 20 s or less
Find the note that mentions a word inside a photo	Both	Correct result opened	3 actions, 10 s or less
Remind yourself to collect a fee tonight	Thư	Reminder set on the item	4 taps or fewer
Insert a formula without typing code	Thư	Formula inserted correctly	0 errors
Write the formula in source mode and export .md	Đức Anh	Valid .md with LaTeX	60 s or less
You deleted a note by mistake. Get it back	Both	Note restored	2 taps or fewer

Heuristic self-evaluation

This is our own review against Nielsen’s 10 heuristics, not a result from user testing. Severity: 0 none, 1 cosmetic, 2 minor, 3 major.

Heuristic	Evidence in the design	Severity
1. Visibility of system status	Sync indicator, offline and conflict banners, snackbars, loading skeleton	0
2. Match with the real world	Paper grid, Trash, wording such as collected or not collected	0
3. User control and freedom	5-second undo, 30-day trash, non-destructive version restore	0
4. Consistency and standards	Shared components, one name per action, same sheet structure	0
5. Error prevention	Password check on leaving the field, confirmation for destructive actions, flagged OCR words	0
6. Recognition over recall	Suggestions for [[, \ and #, visible filter chips, recent searches	0
7. Flexibility and efficiency	Markdown and LaTeX for experts, buttons for beginners. Keyboard shortcuts exist on web only	1
8. Minimalist design	Flat surfaces, one violet action per area, secondary actions in ⋯ and ☰	0
9. Help with errors	Specific messages with an action, such as retry or clear filters	0
10. Help and documentation	Guided tour, sample note, help screen, syntax cheat sheet	0

Part 2: Prototype

I. Mobile

Prototype link: Open the mobile prototype

Figma file: Page 04 · UI Prototype

The prototype has 297 screens in each of the Dark and Light themes. Named flows in the Figma side panel start from sign-in, sign-up, the guided tour and Home. The sun icon on Home switches theme.

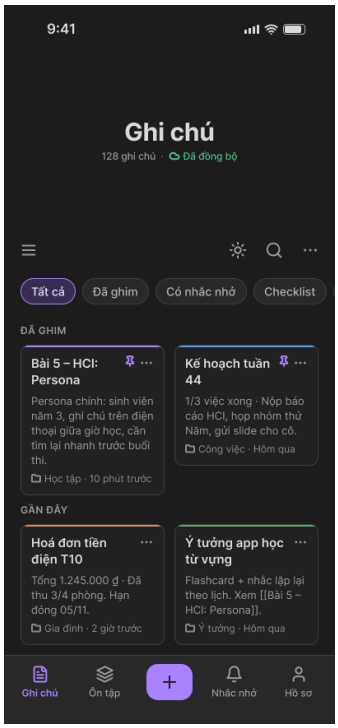
Visual design

- Color: charcoal neutrals with one violet accent, plus semantic colors for success, warning and error. All colors come from shared tokens for Dark and Light.
- Type and icons: Inter, with line icons from the Lucide set.
- Resources: the layout language is inspired by the Obsidian UI kit. No other screens or content were copied.

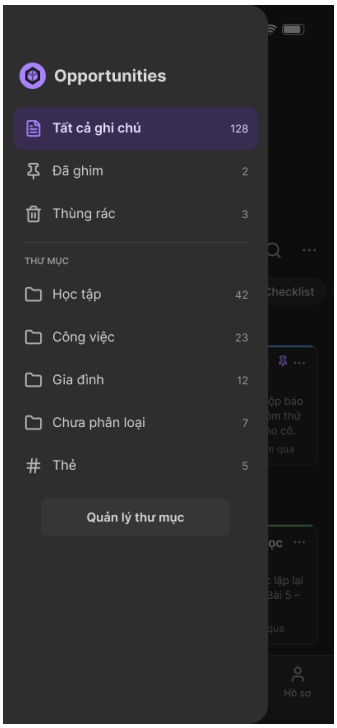
App map

Section	Content	Screens
00	Sign-in and first run	3
01	Entry and browse	28
01b	First-run guided tour	18
01c	Side menu ☰	5
02	Create, write and organize	30
02b	New note by type	22
02c	Editor mode dropdown	3
03	Capture files, handwriting and voice	14
04	Search, folders and tags	17
04b	Folder actions	40
05	Tasks, reminders and notifications	9
05b	Reminder details	20
06	Share and export	13
07	Sync, conflict and version recovery	8
08	Delete, undo and restore	10
09	Profile, settings and writing comfort	21
10	Templates	10
10b	Built-in and saved templates	14
11	Flashcard review	5
12	Smart folders and scan from device	7
Total	**20 sections**	**297**

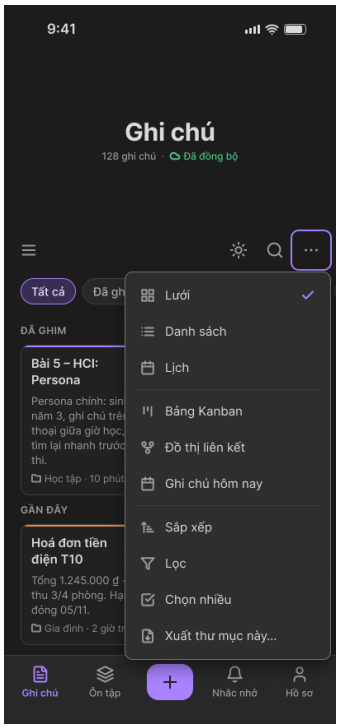
Key screens



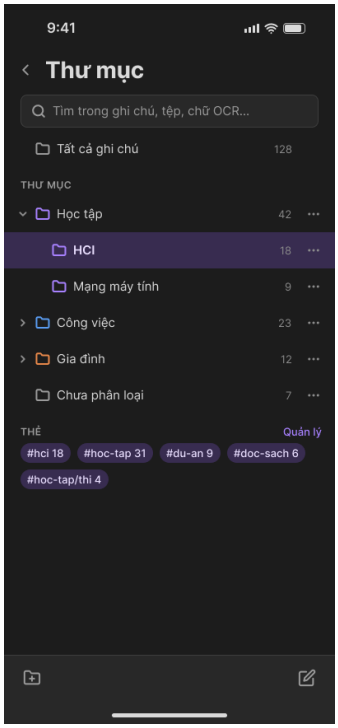
1.1 Home



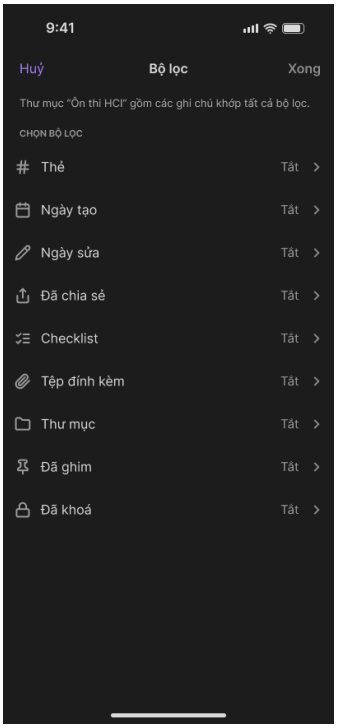
1.28 Side menu



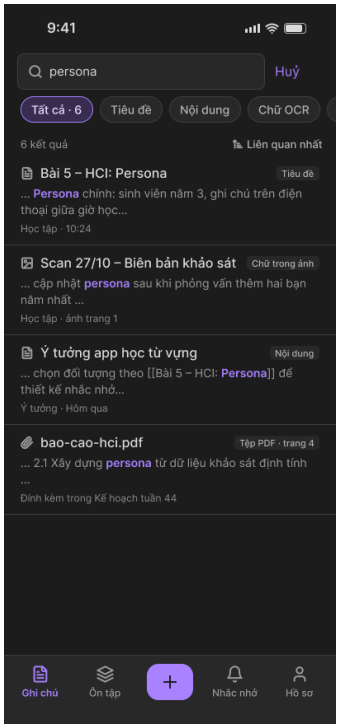
1.6 ··· menu



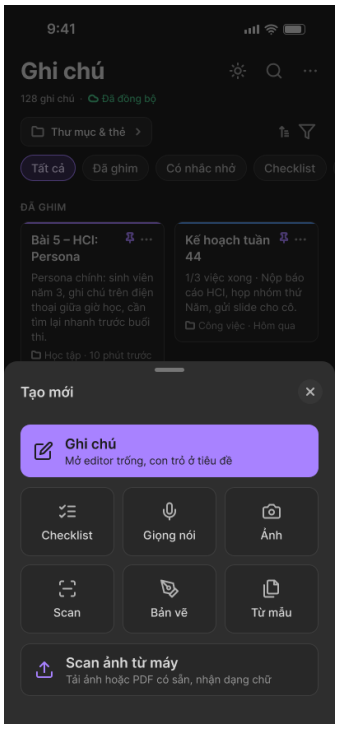
4.4 Folders



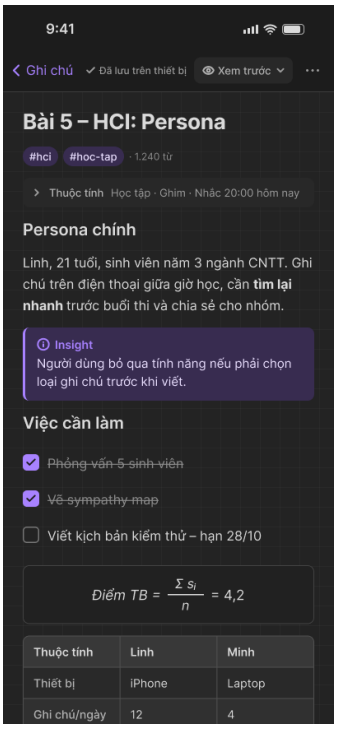
12.1 Smart folder filters



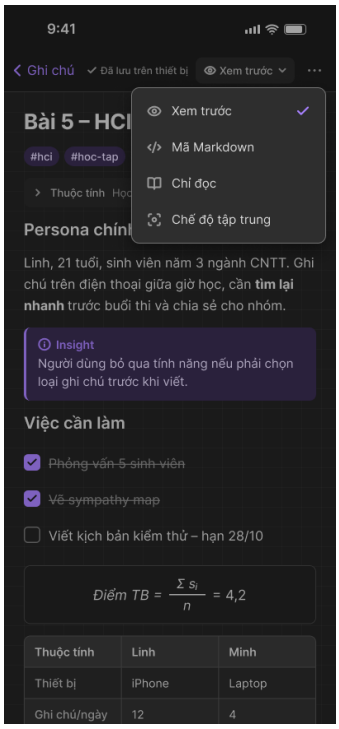
4.2 Search



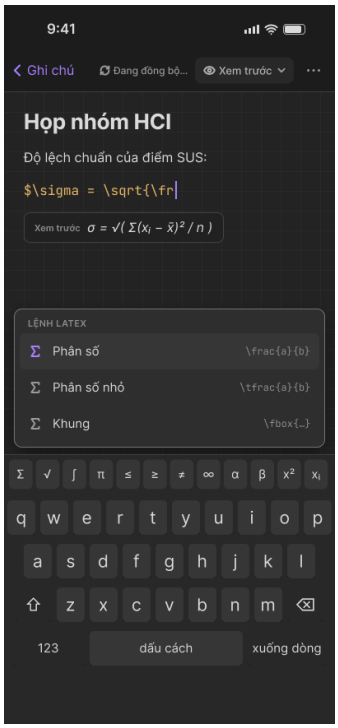
2.1 Create



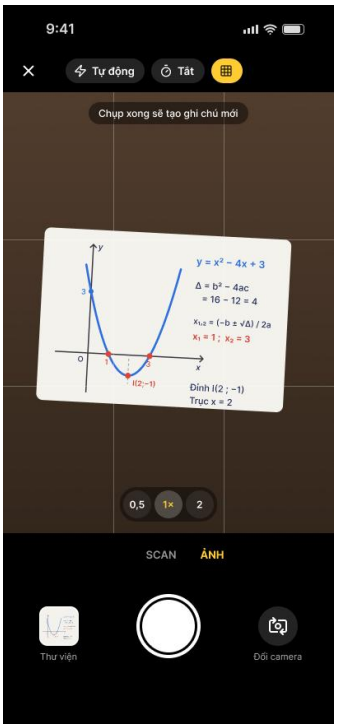
2.3 Editor



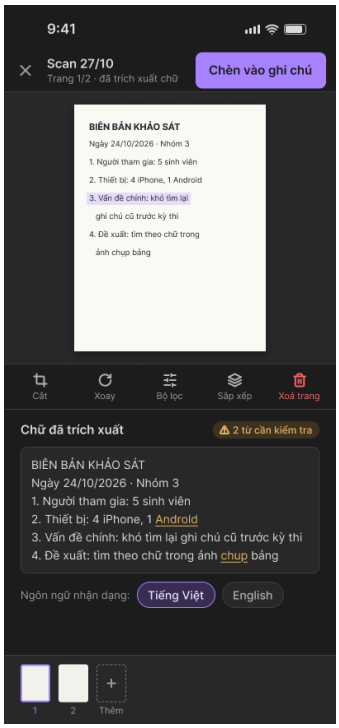
2.31 Mode dropdown



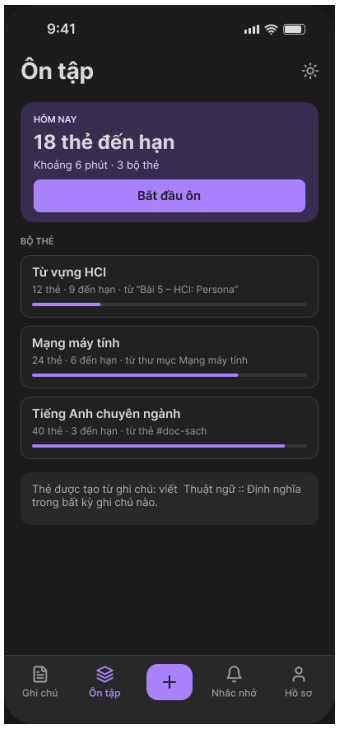
2.8 Formula



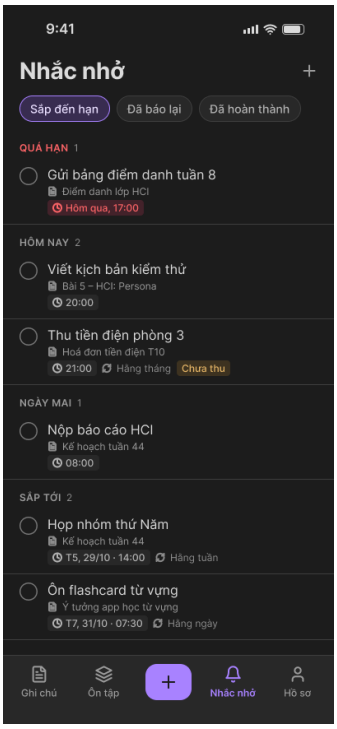
2b.5 Camera



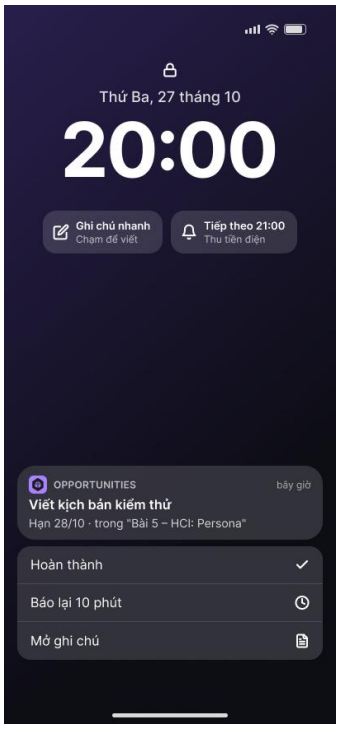
3.4 Text review



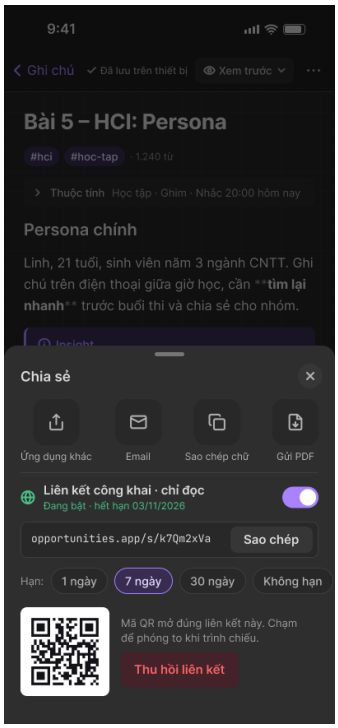
11.1 Review



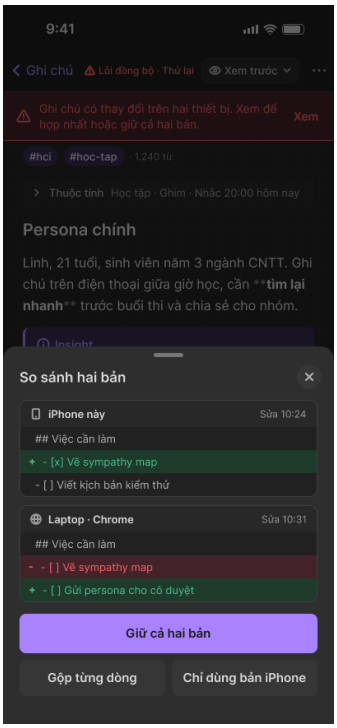
5.1 Reminders



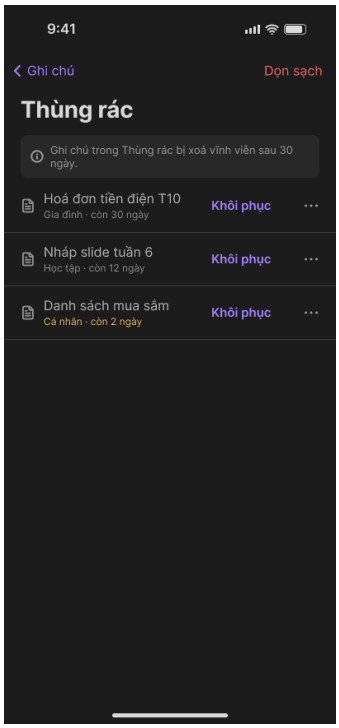
5.3 Notification



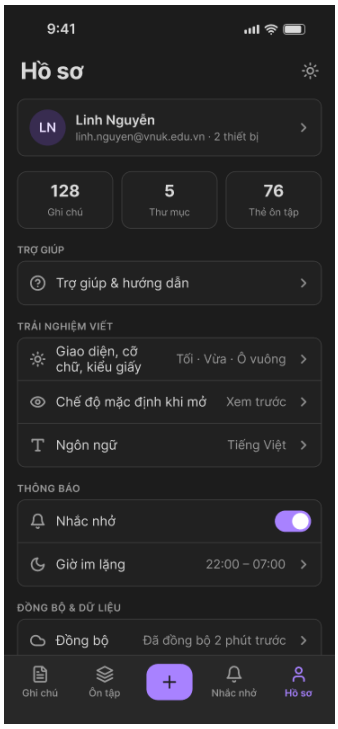
6.2 Share



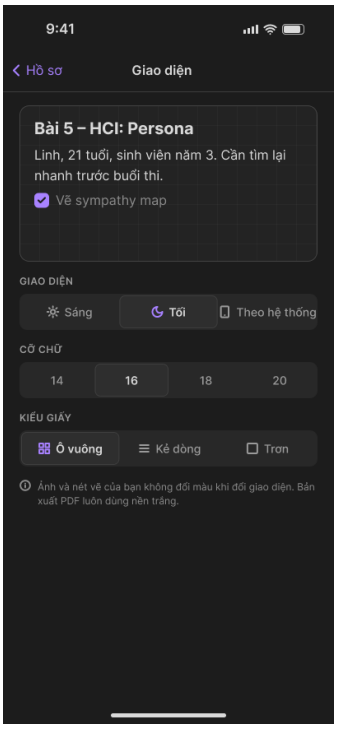
7.2 Conflict



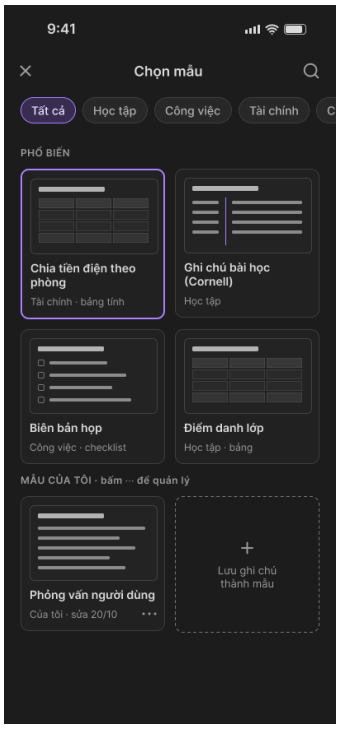
8.2 Trash



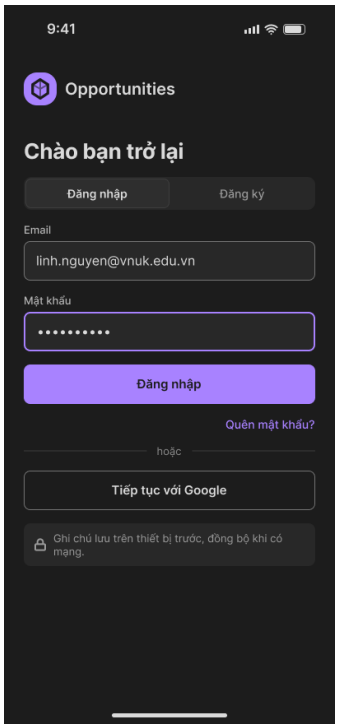
9.1 Profile



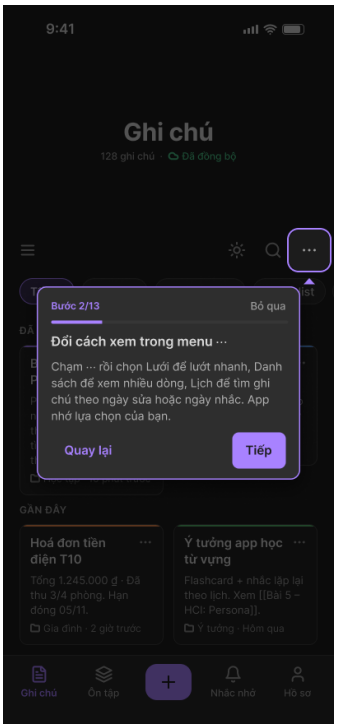
9.2 Appearance



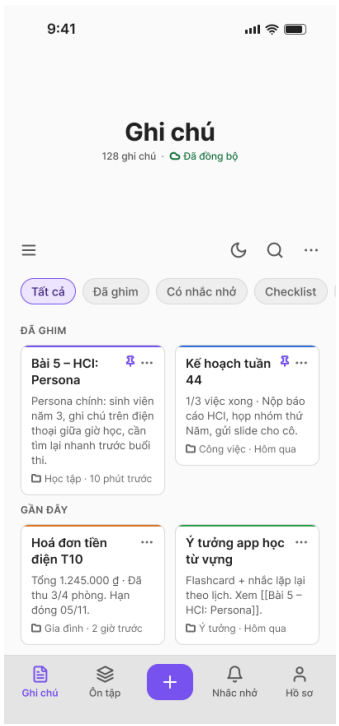
10.1 Templates



A.1 Sign in



T3 Guided tour



1.1 Home, Light theme

Changes after lecturer review

Lecturer feedback	What changed
Move Grid, List and Calendar into the ⋯ button on Home	The switcher is gone from Home. The views are the first items in ⋯, with the current one ticked.
Replace the Search tab with flashcard review	New Review tab with today's cards, decks, a session and a summary. Search stays on Home.
Move folders and tags to Home; Profile takes the freed tab; remove account management and trash from the list; rename the rest	New Profile tab holding all settings on one scrolling screen. Attachments, Templates and Archive sit under Other settings. Trash moved to the side menu.
Recolor the lock icon to match the palette	The lock dialog now uses the violet accent instead of orange.
Put folder, note and tag management on one screen, like Apple Notes	New Folders screen with search, all notes, folders, tags and a bottom toolbar, plus smart folders with filters.
Add scanning of uploaded images when adding a note	New Scan from device option in the Create sheet: pick an image or PDF, review the text, create the note.

We also refined the design after this review. The Home header now follows the spacing of Samsung Notes, with a side menu ☰. The four editor modes moved from ⋯ into one dropdown on the mode box.

II. Web

Prototype link: Open the web prototype

The web app is designed for 1440 px desktop screens, in Light and Dark. It uses a sidebar instead of a bottom bar, and adds keyboard shortcuts and split view. Its navigation was not changed in version 6, so aligning it with the updated mobile navigation is the next step.

Main flows in the web prototype:

- Browse and search notes in Grid, List and Calendar
- Edit and view a note, write in Focus mode
- PDF, voice and drawing inside one note; scan and text recognition
- Template to invoice to PDF export; checklist to reminder; link graph
- Version history, offline sync and conflict resolution
- Folders and tags, sharing with link and QR, archive and restore, account and settings

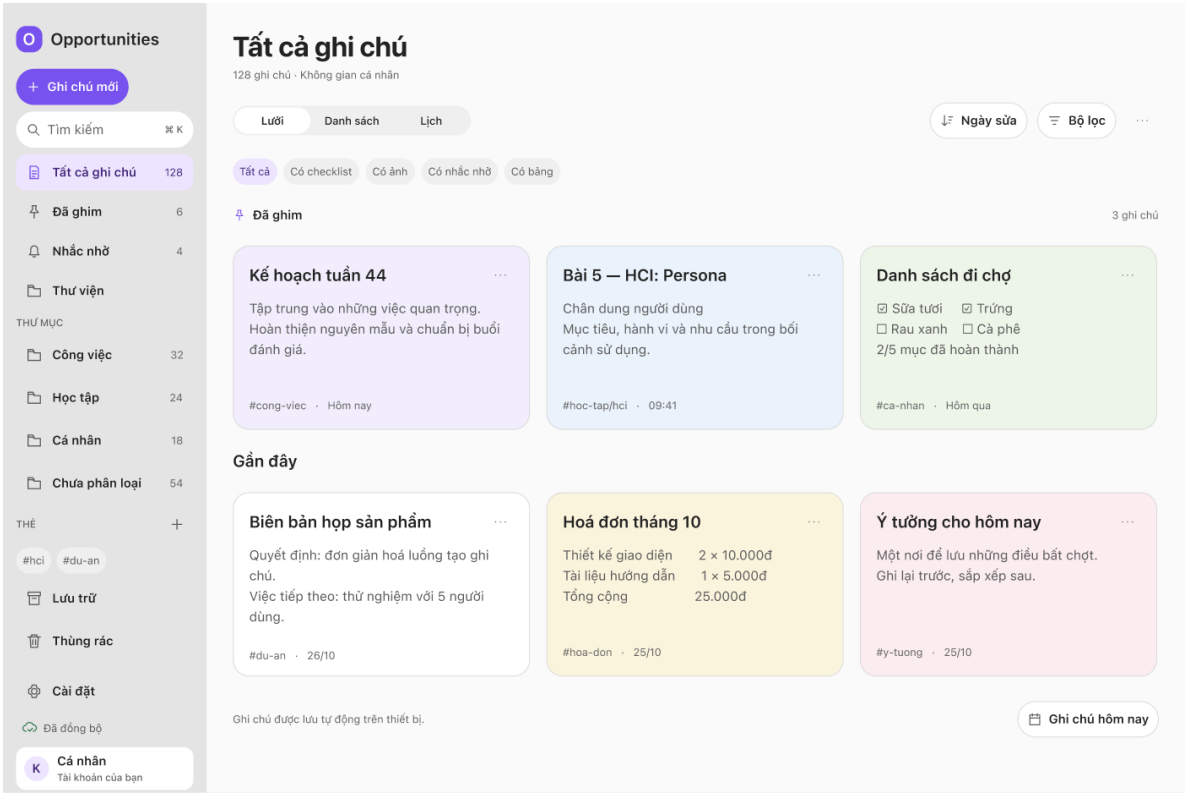


Figure 12. W01 All notes, Grid view

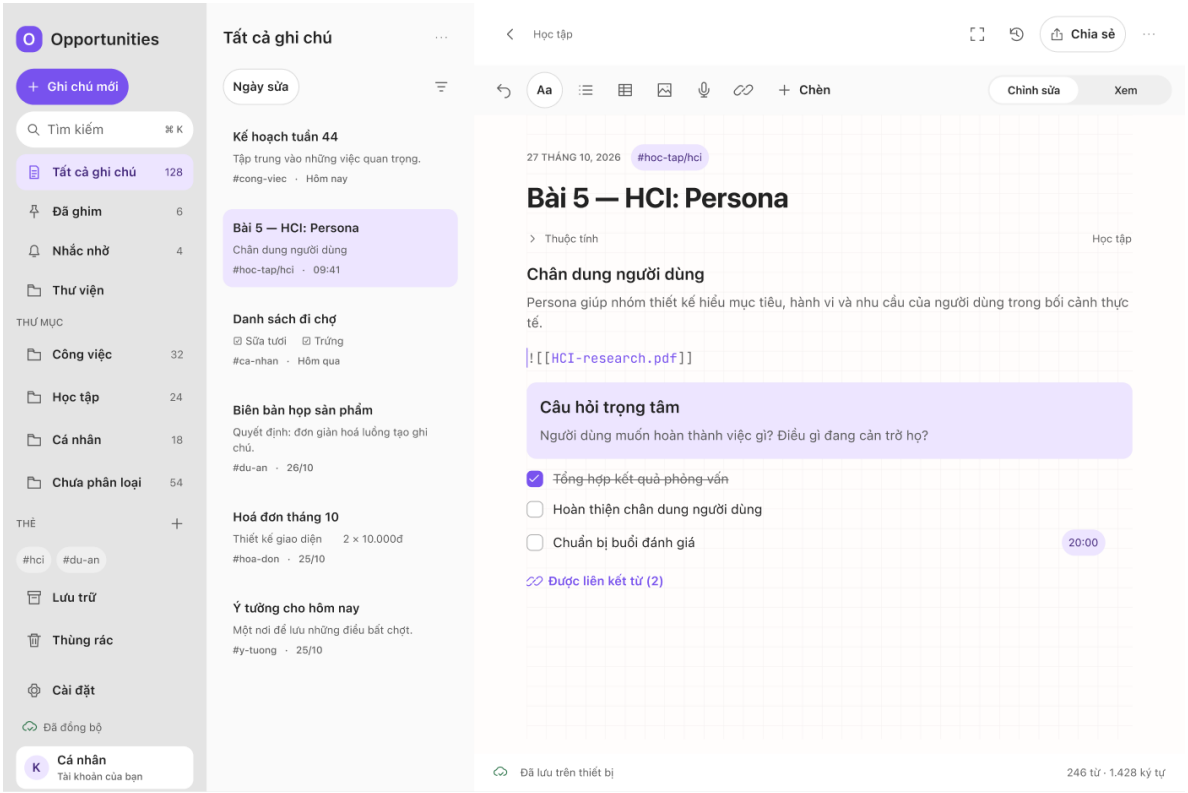


Figure 13. W04 Editor

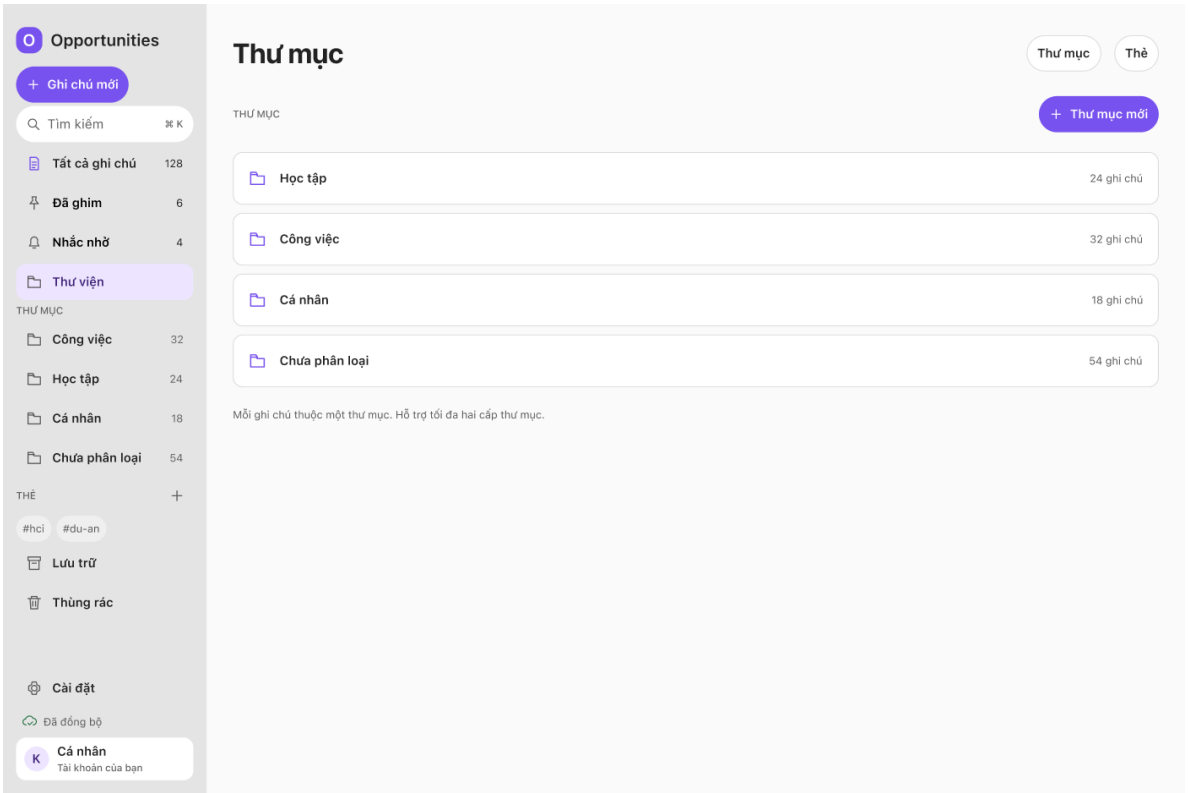


Figure 14. F9.01 Folders

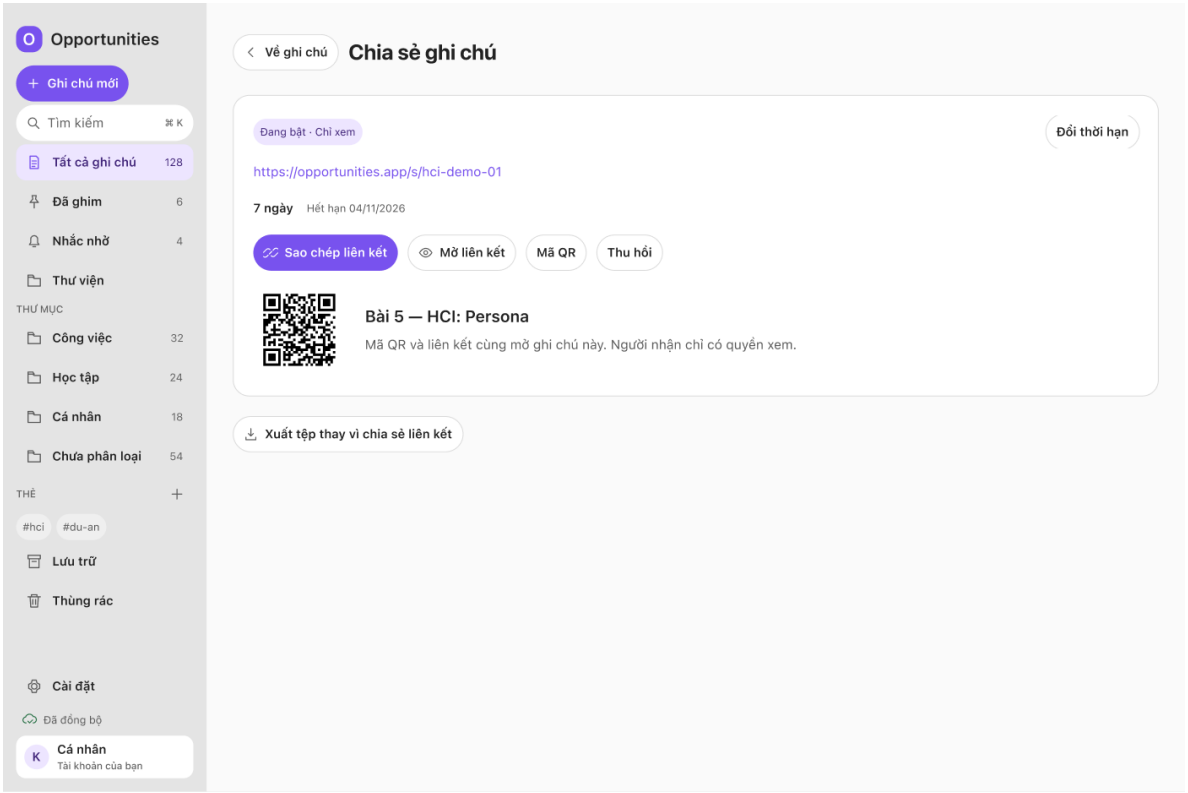


Figure 15. F11.03 Share link created